Randomisation in Design

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Introduction

Randomisation is the protective heart of experimental design. It ensures that in expectation, unknown sources of variation are balanced across treatments; it validates the model's assumption of independent errors; and it prevents systematic bias from lurking variables.

Prerequisites

Experimental design basics.

Theory

Three Fisher principles of DoE: replication, randomisation, blocking. Randomisation: - Ensures **expected balance** of known and unknown covariates. - Justifies the random-error model used in ANOVA. - Validates inference (significance tests are approximate randomisation tests).

Randomisation procedures vary by design: CRD = unrestricted, RCBD = within-block, Latin square = row-column, split-plot = nested.

Assumptions

Randomisation is genuinely random (computer-generated sequence, sealed envelopes); allocation is concealed from the experimenter until assignment.

R Implementation

```r
set.seed(2026)

# Randomise 12 units to 4 treatments for a CRD
treatments <- rep(c("A", "B", "C", "D"), 3)
allocation <- sample(treatments, 12, replace = FALSE)
allocation

# For a RCBD, randomise within each of 3 blocks
blocks <- rep(1:3, each = 4)
alloc_rcbd <- do.call(rbind, lapply(split(1:12, blocks), function(idx) {
  data.frame(unit = idx, treatment = sample(c("A", "B", "C", "D")))
}))
alloc_rcbd
```

Output & Results

A randomised allocation list; randomisation is within block for RCBD to preserve block structure.

Interpretation

"Experimental units were randomly allocated to 4 treatments using a computer-generated sequence (seed 2026); in the RCBD, randomisation was within each block to preserve the local-control benefit."

Practical Tips

- Document the randomisation seed and procedure; reproducibility is part of responsible reporting.
- Conceal allocation from experimenters until assignment; otherwise bias can creep in.
- Randomise run order as well as treatment allocation to guard against temporal confounding.
- Restricted randomisation (within blocks, strata) is required for blocked designs; unrestricted randomisation can mismatch the analysis.
- Randomisation tests (permutation inference) are a rigorous alternative to parametric tests when distributional assumptions fail.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans